

Synthetic Channel Assessment

Compare redundant observations without overstating evidence

Eight fixed learner-operated channels; no external load

Lesson	028 — component	Time	180–240 minutes
Prerequisites	Lessons 001, 002, 007	Board	Arduino Mega 2560
Evidence	eight LED cadences; TP28 and TP29	Status	host verified; bench open
Energy class	E1: USB, passive buttons, resistor-limited LEDs		

Boundary. Every observation is synthetic and selected by the learner. “Short-simulated” is a switch position, not an electrical diagnosis. This circuit contains no relay, launcher, initiator, external load, radio, captured protocol, or generic actuation sink.

1 Mission and model

The lesson preserves a short, inspectable chain:

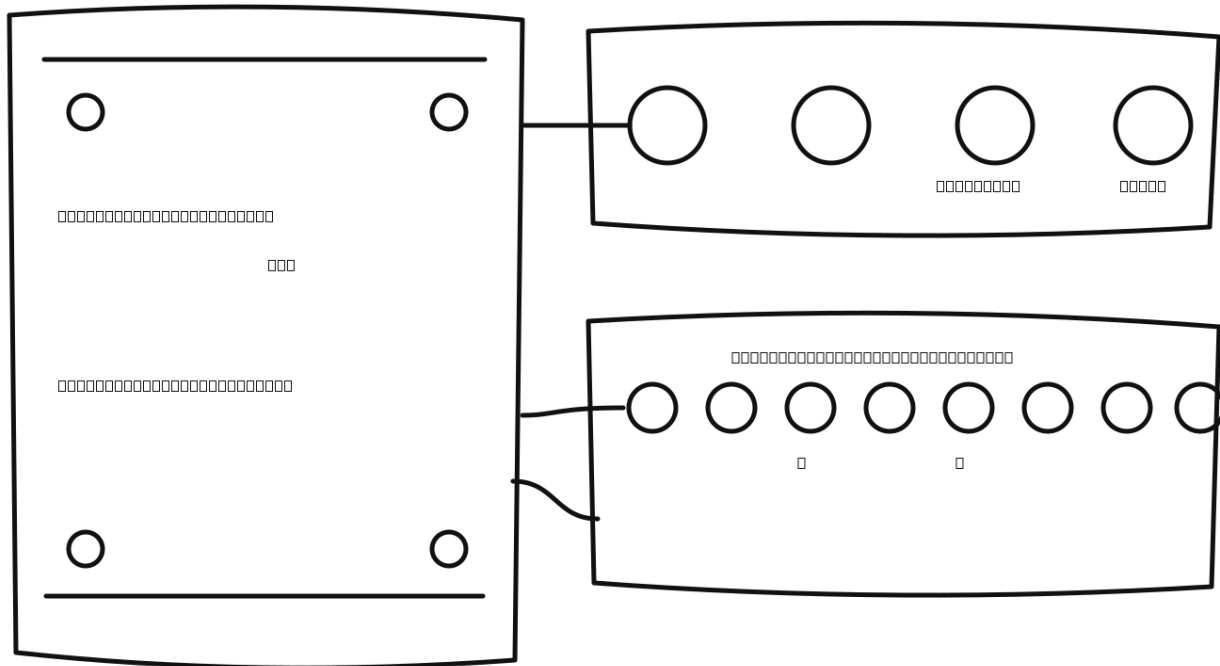
primary + redundant + local age → assessment → LED cadence.

An assessment records what the supplied model says. It does not establish physical continuity, safety, provenance, or permission to energize anything. The core returns values; only the example maps those values to low-voltage indicators.

1.1 Learning outcomes

- derive a state from two explicit observations and local age;
- explain why unavailable, contradictory, and stale are distinct;
- submit a complete set transactionally;
- replay immutable timestamped sets without a catch-up burst;
- separate resource-acquisition evidence from safe-state evidence;
- state precisely what this simulation cannot prove.

2 Orientation and unpowered inspection



Alternative text: graphite orientation plate showing one USB-powered Mega, four pull-up buttons on D30–D33, eight separately resistor-limited LEDs on D22–D29, input test point TP28, and indicator test point TP29. No external-load interface is present.

Role	Mega pin	Connection
channel indicators	D22–D29	pin to 1 k Ω , LED anode; cathode to GND
channel, primary, redundant, apply	D30–D33	momentary switch to GND; internal pull-up
TP28	selected input pin	meter or logic probe relative to Mega GND
TP29	selected indicator pin	meter or logic probe relative to Mega GND

Unpowered prediction. Check one resistor per LED, LED polarity, switch-to-ground wiring, and common ground. Confirm nothing leaves the USB-powered low-voltage teaching circuit. Record measured resistor values: _____.

3 Truth table before code

Primary	Redundant	Assessment	Why
open	open	open	same usable value
closed	closed	closed	same usable value
short-simulated	short-simulated	short-simulated	same synthetic selection
two different usable values	—	contradictory	evidence disagrees
either unavailable	any	unavailable	one required observation is absent
either older than threshold	any	stale	age dominates classification

Complete the predictions:

Input	Predicted state	Observed cadence	Interpretation
open / closed	_____	_____	_____
unavailable / closed	_____	_____	_____
closed / closed at 5000 ms	_____	_____	_____
closed / closed at 5001 ms	_____	_____	_____

Age equal to **staleAfter** retains the underlying state. The first greater age is stale. The half-range timestamp rule prevents an ambiguous future value from being mistaken for old evidence.

4 Lifecycle and transactional update

Construction is inert. `initialize()` rejects a zero or ambiguous stale interval, repeated initialization is harmless, `shutdown()` is idempotent, and destruction calls shutdown. No heap, exception, RTTI, hidden clock, callback, or hardware owner appears in the assessor.

`update()` first validates the complete submission. It rejects:

- more than eight entries or null storage with a nonzero count;
- an ID outside 0–7 or the same ID twice;
- an invalid observation enum;
- a timestamp in the ambiguous future half of the 32-bit space.

Only after all entries pass does it replace slots. Therefore a bad final entry cannot partly overwrite earlier channels.

4.1 Transaction worksheet

Start with channel 0 closed and channel 1 open. Submit channel 0 as open, then include an invalid duplicate channel 1.

1. Expected update status: _____
2. Channel 0 after rejection: _____
3. Why must validation precede mutation? _____
4. Does omitting channel 7 erase its prior observation? _____

5 Recorded evidence source

`RecordedInertObservationSource` walks caller-owned immutable sets. Each set holds all eight observations and one due time. Times are strictly increasing and remain within the unambiguous 32-bit interval.

```
source.update    (now);
snapshot = source.snapshot ();
assessor.update (now, snapshot.observations,
                  snapshot.observationCount);
```

A delayed update selects the latest due complete set. It never emits each missed set as a burst. Repeated snapshots expose the same pointers and values until another set becomes due. The caller must keep the immutable storage alive longer than the source.

5.1 Replay exercise

Sets are due at 100, 200, and 300 ms. Calls arrive at 90, 250, 250, and 400 ms. Write the selected due time after each call: _____, _____, _____, _____.

Explain why the 100 ms set is not surfaced at 250 ms:

_____.

6 Narrative Mega example

```
observeOperator    (now);
decideAssessments (now);
showAssessments   (now);
```

Objects appear in dependency order: runtime, buttons, indicators, assessor, then observation values. Setup acquires every owner, configures all channels open, and starts the panel. The loop observes debounced events, decides only when Apply submits a complete set, then presents all assessments.

Channel selects 0–7. Primary and Redundant cycle open, closed, short-simulated, unavailable. Apply timestamps and submits the selected channel while preserving the other seven entries.

State	Indicator	Interpretation
open	off	matching open selections
closed	steady	matching closed selections
short-simulated	1 s on / 1 s off	matching synthetic short selection
stale	200 ms pulse each second	last accepted observation is too old
contradictory	200 ms alternating	usable observations disagree
unavailable	two 120 ms pulses per 2 s	one observation unavailable

7 Predict, observe, interpret

1. Predict Open/Open leaves the selected indicator off. Apply it, observe TP29, and interpret only against the model.
2. Predict Closed/Closed is steady. Apply and compare the LED with TP29.
3. Predict Open/Closed produces the fast contradictory cadence.
4. Predict either Unavailable selection produces two pulses.
5. Stop applying. Predict and observe stale after five seconds.
6. Press Channel, change another pair, and verify the first channel keeps its prior accepted observation until it ages.

Two separate checks. A successful startup response supports the claim that resources were acquired. It does not prove shutdown. Invoke shutdown, then measure TP29 for high impedance; a dark LED is insufficient.

8 Diagnosis

Observation	Likely cause	Next bounded check
no indicators ever respond	acquisition failure or wiring error	inspect D22–D29 and each resistor unpowered
button never changes selection	switch not reaching GND	observe TP28 high then low
unexpected contradictory cadence	selected values differ	cycle both banks to a known matching pair
all active channels become stale	no recent Apply event	apply one known pair and time five seconds
one LED disagrees with TP29	LED polarity or open resistor path	compare pin level, resistor, and diode orientation
shutdown LED dark but pin driven	light is not high-impedance proof	measure TP29 electrically

9 Exercises

1. Derive all sixteen primary/redundant pairs before running the sketch.
2. Add a host trace around UINT32_MAX that crosses the stale boundary without changing its elapsed result.
3. Prove a duplicate channel in the final array position leaves every previous slot unchanged.
4. Explain why a callback named for a generic output would violate the inert boundary even if the canonical sketch used LEDs.
5. Design a presentation adapter that accepts assessment values but owns no source and cannot mutate the assessor.

10 CLI and acceptance record

```
make host-test
make arduino-Lesson028InertChannelAssessment
make size-Lesson028InertChannelAssessment
make upload-Lesson028InertChannelAssessment PORT=/dev/ttyACM0
make serial-log PORT=/dev/ttyACM0
make lessons
make lessons-check
make site-check
make hardware-card LESSON=028 BOARD=mega2560
```

Serial is optional supporting evidence and the sketch emits none. Compilation and host replay do not close this physical record.

Board and revision	
Exact buttons, LEDs, resistors	
Supply and instrument	
Tool versions and commit	
Unpowered inspection	
Acquisition prediction / observation	
Six state predictions / observations	
TP28 and TP29 readings	
Shutdown high-impedance evidence	
Reset and power-removal observation	
Deviation and disposition	

Status after software gates: host verified; hardware acceptance open.

11 Reference trail

- Searchable Lesson 028 HTML reference
- Public assessment and recorded-source interface
- Deterministic lifecycle, transaction, timing, and replay tests
- Lessons 028–030 safety and composition design
- Arduino Mega 2560 Rev3 datasheet